

TCR($\mathbb{F}_2$) VIA THE SLICE SPECTRAL SEQUENCE

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In this note, we use a lift of the slice filtration to the ∞ -category of spectra with C_2 -twisted $\mathbb{T}$ -action in order to compute the homotopy groups of $\mathrm{TCR}(\mathbb{F}_2)$. This results in a simplification of the computation by Quigley–Shah [QS21].

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1. THE $\mathbb{T}$ -WISTED SLICE FILTRATION

We use the theory of C_2 -spectra with twisted group actions from [QS25, QS21]. Let $\underline{\mathrm{Sp}}$ be the C_2 - ∞ -category of C_2 -spectra. To the extension $0 \rightarrow \mathbb{T} \rightarrow O(2) \rightarrow C_2 \rightarrow 0$, we can associate the ∞ -category of C_2 -spectra with $\mathbb{T}$ -wisted action $\mathrm{Sp}^{h_{C_2}\mathbb{T}} := \mathrm{Fun}_{C_2}(B_{C_2}\mathbb{T}, \underline{\mathrm{Sp}})$.

For any $\mathbb{E}_\sigma$ -ring A , we view its Real topological Hochschild homology $\mathrm{THR}(A)$ as an object of $\mathrm{Sp}^{h_{C_2}\mathbb{T}}$ equipped with a parametrized Tate-valued Frobenius map

$$\varphi : \mathrm{THR}(A) \rightarrow \mathrm{THR}(A)^{t_{C_2}\mu_2}$$

in $\mathrm{Sp}^{h_{C_2}\mathbb{T}}$. When it comes to C_2 -spectra we mostly follow the notation and conventions of Hill–Hopkins–Ravenel [HHR16, HHR17].

We review the construction of the C_2 -slice filtration in the context of C_2 - ∞ -categories following [Car25]. This perspective allows us to construct the slice filtration in the twisted setting in a straightforward way and allows one to deduce useful multiplicative properties of the filtration.

Definition 1.1. For a C_2 - ∞ -category, $\mathcal{C}$, we let $\mathrm{Fil}(\mathcal{C})$ and $\mathrm{Gr}(\mathcal{C})$ denote the ∞ -categories of C_2 -functors $\mathrm{Fun}_{C_2}(\mathbb{Z}^{\mathrm{op}}, \mathcal{C})$ and $\mathrm{Fun}_{C_2}(\mathbb{Z}^\delta, \mathcal{C})$, respectively. .

For each $n \in \mathbb{Z}$ and subgroup $K \subseteq C_2$, let $\mathrm{Sp}_{\geq n}^K$ be the subcategory of Sp^K generated under colimits and extensions by the collection

$$\{\mathrm{Ind}_H^K S^{k\rho_H} \mid H \subseteq K, k|H| \geq n\}.$$

There results a sequence of full subcategory inclusions

$$\cdots \hookrightarrow \mathrm{Sp}_{\geq n}^K \hookrightarrow \mathrm{Sp}_{\geq n-1}^K \hookrightarrow \cdots \hookrightarrow \mathrm{Sp}^K.$$

For $K = e$, this is evidently the usual Whitehead filtration on Sp . For $K = C_2$, we recover the (regular) C_2 -slice filtration defined in [Ull13].

These two filtrations assemble into an *normed $\mathbb{Z}$ -indexed slice filtration* on $\underline{\mathrm{Sp}}$ in the sense of [Car25, Definition 4.5]. We refer the reader to *op.cit.* for details and simply state the following consequence that we will make use of.

Theorem 1.2 ([Car25, Proposition 4.3], [Car25, Proposition 4.4]). *For each subgroup $K \subseteq C_2$, there is a left and right complete t -structure on $\mathrm{Fil}(\mathrm{Sp}^K)$ such that $X \in \mathrm{Fil}(\mathrm{Sp}^K)$ is connective if and only if $X_n \in \mathrm{Sp}_{\geq n}^K$ for each $n \in \mathbb{Z}$. The assignment*

$$\mathrm{Fil}(\underline{\mathrm{Sp}})_{\geq 0}(G/K) := \mathrm{Fil}(\mathrm{Sp}^K)_{\geq 0}$$

defines a C_2 -symmetric monoidal subcategory of $\mathrm{Fil}(\underline{\mathrm{Sp}})$, and the connective cover functors assemble into a lax C_2 -symmetric monoidal right adjoint

$$\tau_{\geq 0}^{\mathrm{slice}} : \mathrm{Fil}(\underline{\mathrm{Sp}}) \rightarrow \mathrm{Fil}(\underline{\mathrm{Sp}})_{\geq 0}.$$

Definition 1.3. The *parametrized C_2 -slice filtration* is the composite C_2 -functor

$$\underline{P} : \underline{\mathrm{Sp}} \xrightarrow{\mathrm{const}} \mathrm{Fil}(\underline{\mathrm{Sp}}) \xrightarrow{\tau_{\geq 0}^{\mathrm{slice}}} \mathrm{Fil}(\underline{\mathrm{Sp}})_{\geq 0}.$$

The *parametrized $\mathbb{T}$ -wisted C_2 -slice filtration* is the C_2 -functor

$$\hat{\underline{P}} : \mathrm{Sp}^{h_{C_2}\mathbb{T}} \rightarrow \mathrm{Fil}(\mathrm{Sp}^{h_{C_2}\mathbb{T}})$$

defined by postcomposition with $\underline{P}$.

Remark 1.4. There is a forgetful functor $\sigma_{\mathbb{T}}^* : \mathrm{Sp}^{h_{C_2}\mathbb{T}} \rightarrow \mathrm{Sp}^{C_2}$, which carries the twisted slice filtration to the original C_2 -slice filtration: $\sigma_{\mathbb{T}}^*(\hat{P}_{\geq *}) \simeq P_{\geq *} \sigma_{\mathbb{T}}^*(X)$.

Theorem 1.5. *Let $X \in \mathrm{Sp}^{h_{C_2}\mathbb{T}}$ be a C_2 - $\mathbb{E}_{\infty}$ -ring. Then the filtered objects $\hat{P}_{\geq *}X$, $P_{\geq *}X$, $(\hat{P}_{\geq *}X)^{h_{C_2}\mathbb{T}}$, and $(\hat{P}_{\geq *}X)^{t_{C_2}\mathbb{T}}$ are C_2 - $\mathbb{E}_{\infty}$ -rings in their respective categories. Furthermore, the natural maps*

$$(\hat{P}_{\geq *}X)^{h_{C_2}\mathbb{T}} \rightarrow P_{\geq *}X, \quad \text{and} \quad \mathrm{can} : (\hat{P}_{\geq *}X)^{h_{C_2}\mathbb{T}} \rightarrow (\hat{P}_{\geq *}X)^{t_{C_2}\mathbb{T}}$$

are C_2 - $\mathbb{E}_{\infty}$ -ring maps in Sp^{C_2} .

Proof. Theorem 1.2 implies that $\underline{P}$ is a lax C_2 -symmetric monoidal functor since const is. The result then follows from the fact that both $(-)^{t_{C_2}G}$ and $(-)^{t_{C_2}G}$ are lax symmetric monoidal. $\square$

The following propositions show that in many cases we have a good control over the $\mathbb{T}$ -wisted slice associated graded.

Definition 1.6. If $\underline{M}$ is a C_2 -Mackey functor, let $\mathcal{P}^0 \underline{M}$ denote quotient of $\underline{M}$ obtained by killing the kernel of the restriction map $\mathrm{res} : \underline{M}(C_2/C_2) \rightarrow \underline{M}(C_2/e)$.

Proposition 1.7. *For any C_2 -spectrum, X , we have*

$$P_{2k}^{2k} X \simeq \Sigma^{k\rho} H \pi_{k\rho} X,$$

and

$$P_{2k+1}^{2k+1} X \simeq \Sigma^{k\rho+1} H(\mathcal{P}^0 \pi_{k\rho+1} X).$$

Proof. This is stated without proof in [Hil20, Theorem 17.5.25]. However, a proof is readily obtained, for example, by modifying the proof of [AKKQ25, Proposition 2.6] using $\pi_{k\rho+1}$ instead of $\pi_{k\rho-1}$. $\square$

Proposition 1.8. *Suppose $H \in \mathrm{Sp}^{h_{C_2}\mathbb{T}}$ is such that $\sigma_{\mathbb{T}}^*(H) \simeq \Sigma^V H\mathbf{M}$ for a C_2 -Mackey functor $\mathbf{M}$. Then H has trivial $\mathbb{T}$ -wisted action if either of the following holds:*

- (1) *the restriction map of $\mathbf{M}$ is injective*
- (2) *the transfer map of $\mathbf{M}$ is surjective.*

Proof. This is proved in [QS21, Proposition 6.10]. Though it is stated there in the case $\mathbf{M}$ is a constant Mackey functor, that assumption is only used to guarantee that the restriction map is an isomorphism. Their proof holds more generally in either case above. $\square$

2. $\mathrm{TCR}^-(\mathbb{F}_2)$ AND $\mathrm{TPR}(\mathbb{F}_2)$

We now use the $\mathbb{T}$ -wisted slice filtration to construct spectral sequences computing parametrized homotopy fixed points and parametrized Tate construction.

Definition 2.1. The *twisted homotopy fixed point spectral sequence* for $X \in \mathrm{Sp}^{h_{C_2}\mathbb{T}}$ is the spectral sequence of $\mathrm{RO}(C_2)$ -graded C_2 -Mackey functors associated to the filtered C_2 -spectrum

$$(\hat{P}_{\geq *} X)^{h_{C_2}\mathbb{T}}$$

obtained by applying parametrized homotopy fixed points levelwise to the $\mathbb{T}$ -wisted slice tower of X .

Explicitly, we have a spectral sequence with signature

$$E_2^{s,V}(X) = \pi_{V-s}(\hat{P}_{|V|}^{[V]} X)^{h_{C_2}\mathbb{T}} \implies \pi_{V-s} X^{h_{C_2}\mathbb{T}},$$

and differentials

$$d_r : E_r^{s,V} \rightarrow E_r^{s+r, V+r-1}.$$

Example 2.2. Let $X = H\mathbf{M}$ with $\mathbf{M}$ a Green functor satisfying any of the conditions of Proposition 1.8. Then the twisted homotopy fixed point spectral sequence collapses at the E_2 page to give

$$\pi_{\star} H\mathbf{M}^{h_{C_2}\mathbb{T}} \cong \underline{H}^{-\star}(B_{C_2}\mathbb{T}; \mathbf{M}) \cong \mathbf{M}[\bar{y}].$$

Now suppose additionally that $\pi_{\star\rho-1}^{C_2} H\mathbf{M} = 0$. Then arguing as in the proof of [QS21, Proposition 6.14], we also have

$$\pi_{\star} H\mathbf{M}^{t_{C_2}\mathbb{T}} \cong \mathbf{M}[\bar{y}^{\pm}],$$

with $|\bar{y}| = -\rho$.

With this spectral sequence at hand, we can compute Real negative cyclic homology of $\mathbb{F}_2$. We find that the spectral sequence is entirely determined by the fact that the underlying spectral sequence collapses and the existence of a multiplicative extension on the underlying E_{∞} -page.

Theorem 2.3. The $\mathrm{RO}(C_2)$ -graded homotopy Green functor of Real topological negative cyclic homology of $\mathbb{F}_2$ is given by

$$\pi_{\star} \mathrm{TCR}^-(\mathbb{F}_2) \cong \frac{H\underline{\mathbb{Z}}_{2\star}[\bar{x}, \bar{y}]}{(\bar{x}\bar{y} - 2)},$$

where $|\bar{x}| = \rho$ and $|\bar{y}| = -\rho$.

Proof. We consider $\mathrm{THR}(\mathbb{F}_2)$, which in [DMPR21] was found to be equivalent to a sum

$$\mathrm{THR}(\mathbb{F}_2) \simeq \bigoplus_{k \geq 0} \Sigma^{k\rho} H\mathbb{F}_2.$$

Thus $P_k^k \mathrm{THR}(\mathbb{F}_2) \simeq \Sigma^{k\rho} H\mathbb{F}_2$ and by Proposition 1.8, the twisted HFPSS has signature

$$\underline{H}_*[\bar{x}, \bar{y}] \implies \pi_* \mathrm{TCR}^-(\mathbb{F}_2),$$

with $|\bar{x}| = (\rho, 0)$, $|\bar{y}| = (-\rho, 2)$. For degree reasons $\bar{x}$, and $\bar{y}$ must be permanent cycles. It remains to determine the fate of a_σ , u_σ , and the negative cone.

The only possible targets of differentials on a_σ are of the form $\bar{y}^{i+1} \bar{x}^i$. Their restrictions are nonzero surviving classes in the underlying spectral sequence, i.e. the usual homotopy fixed point spectral sequence computing the homotopy of $\mathrm{TC}^-(\mathbb{F}_2)$. It follows that a_σ must be a permanent cycle.

Now, the underlying spectral sequence degenerates at E_2 with no differentials. This automatically implies that any class which is a transfer on E_2 , in particular θ/u_σ^k for $k \geq 0$, must be a permanent cycle.

There are extensions on the underlying E_∞ -page coming from $xy = 2$. By restriction, this implies that $\bar{x}\bar{y}$ must survive and be a hidden 2-extension on the unit: $\bar{x}\bar{y} = 2$. Considering the resulting homotopy Mackey functor extension, this implies that $\bar{x}\bar{y}$ is a transfer in homotopy. It then follows from [HHR17, Theorem 4.4] that $a_\sigma \bar{x}\bar{y}$ must be killed by a differential in the spectral sequence. The only possibility is

$$(2.0.1) \quad d_3 u_\sigma = a_\sigma \bar{x}\bar{y}.$$

Now observe that by the Leibniz rule and since $u_\sigma \theta = 0$,

$$0 = d_3(u_\sigma \frac{\theta}{a_\sigma}) = \bar{x}\bar{y}\theta + u_\sigma d_3(\frac{\theta}{a_\sigma}).$$

Thus, we must have

$$(2.0.2) \quad d_3(\frac{\theta}{a_\sigma}) = \bar{x}\bar{y} \frac{\theta}{u_\sigma}.$$

We note that the only other possibility would be that this differential hits

$$\bar{x}\bar{y} \frac{\theta}{u_\sigma} + \bar{x}^2 \bar{y}^3 \frac{\theta}{a_{2\sigma}}.$$

However, this would imply a nontrivial differential on θ , which has already been ruled out.

The differentials in this spectral sequence are then determined by Eq. (2.0.1) and Eq. (2.0.2) under the Leibniz rule. In particular, the differentials on the negative cone are given by

$$d_3(\frac{\theta}{a_\sigma^k u_\sigma^\ell}) = \begin{cases} \bar{x}\bar{y} \frac{\theta}{a_\sigma^{k-1} u_\sigma^{\ell+1}} & k \geq 1, \ell \text{ even} \\ 0 & \text{else.} \end{cases}$$

There are no further possible differentials. □

Remark 2.4. Explicitly, at the fixed point level we have

$$\pi_*^{C_2} \mathrm{TCR}^-(\mathbb{F}_2) \cong \frac{\mathbb{Z}_2[u_\sigma^2, a_\sigma, \bar{x}, \bar{y}]}{(2a_\sigma, \bar{x}\bar{y} - 2)} \oplus \mathbb{Z}_2\{\frac{\theta \bar{x}^a \bar{y}^b}{u_\sigma^{2\ell}} \mid \ell \geq 0\} \oplus \mathbb{F}_2\{\frac{\theta \bar{x}^a \bar{y}^b}{a_\sigma^k u_\sigma^{2j+1}} \mid j, k, a, b \geq 0\},$$

where it is understood that $\theta\bar{x}\bar{y} = 2\theta$, so in particular

$$\frac{\theta\bar{x}^a\bar{y}^b}{a_\sigma^k u_\sigma^{2j+1}} = 0 \quad \text{if } a \neq 0 \text{ and } b \neq 0.$$

Definition 2.5. The *twisted Tate spectral sequence* for $X \in \mathrm{Sp}^{h_{C_2}\mathbb{T}}$ is the spectral sequence associated to the filtered C_2 -spectrum

$$(\hat{P}_{\geq *}X)^{t_{C_2}\mathbb{T}}$$

obtained by applying the parametrized Tate construction levelwise to the twisted slice tower of X .

Theorem 2.6. The $\mathrm{RO}(C_2)$ -graded homotopy Green functor of Real topological periodic homology of $\mathbb{F}_2$ is given by

$$\pi_* \mathrm{TPR}(\mathbb{F}_2) \cong H\underline{\mathbb{Z}}_*[\bar{y}^\pm],$$

where $|\bar{y}| = -\rho$. Furthermore, the canonical map

$$\mathrm{can} : \pi_* \mathrm{TCR}^-(\mathbb{F}_2) \rightarrow \pi_* \mathrm{TPR}(\mathbb{F}_2)$$

is given by inverting $\bar{y}$.

Proof. The twisted TateSS for $\mathrm{THR}(\mathbb{F}_2)$ has signature

$$\underline{H}_*[\bar{x}, \bar{y}^\pm] \implies \pi_* \mathrm{TPR}(\mathbb{F}_2).$$

The map from the twisted HFPSS on E_2 -pages is given by inverting $\bar{y}$. It follows that $\bar{x}, \bar{y}^\pm, a_\sigma, \theta/u_\sigma^k$ are all permanent cycles, being the images of permanent cycles.

Likewise, the class $a_\sigma\bar{x}\bar{y}$ must be killed by a $d_{\leq 3}$, and the only possibility is that the differential Eq. (2.0.1) happens in the TateSS as well. The differential Eq. (2.0.2) then follows. $\square$

Remark 2.7. The reader may wish to compare the last two examples to the very similar computations in [QS21, Section 6.2]. The E_2 -pages obtained here are the same as those in *op.cit.* up to a reindexing. In both cases, the differentials are generated by under the Leibniz rule by a differential of the form Eq. (2.0.1). The key difference is that we simplify the argument for obtaining this differential using knowledge of hidden extensions in the underlying spectral sequence and the Mackey functor structure.

3. THE FROBENIUS AND $\mathrm{TCR}(\mathbb{F}_2)$

Proposition 3.1. *There is an equivalence*

$$\mathrm{THR}(\mathbb{F}_2)^{t_{C_2}\mu_2} \simeq H\underline{\mathbb{F}}_2[b^\pm],$$

with $|b| = -\rho$. Furthermore, under this identification, the Tate-valued Frobenius φ can be identified with the $\mathbb{T}$ -wisted slice connective cover

$$\varphi : \mathrm{THR}(\mathbb{F}_2) \rightarrow \mathrm{THR}(\mathbb{F}_2)^{t_{C_2}\mu_2}$$

Proof. We use the μ_2 -twisted slice spectral sequence associated to $(\hat{P}_{\geq *} \mathrm{THR}(\mathbb{F}_2))^{t_{C_2}\mu_2}$. We have the E_2 -page given by a polynomial algebra over the Tate cohomology of $B_{C_2}\mu_2$ [HK01]

$$H\underline{\mathbb{F}}_2^{t_{C_2}\mu_2}[\bar{x}] \cong \hat{H}^*(B_{C_2}\mu_2; \mathbb{F}_2)[\bar{x}] \cong \frac{H_*[c, b^\pm][\bar{x}]}{c^2 = a_\sigma c + u_\sigma b}.$$

The restriction map from the $\mathbb{T}$ -twisted spectral sequence implies the differential

$$d_3 u_\sigma = a_\sigma \bar{x} b.$$

The relation $c^2 = a_\sigma c + u_\sigma b$ together with the Leibniz rule then imply the differential

$$d_3 c = b^2 \bar{x}.$$

The remaining E_2 -page generators are permanent cycles since they are images of permanent cycles in the $\mathbb{T}$ -twisted spectral sequence. The first of these differentials implies an extension $\bar{x}b = 2$ in homotopy. The second of these then implies $2b = 0$, thus proving the first part of the statement.

To determine φ on homotopy, we use the Mackey functor structure. On the underlying level, the Tate-valued Frobenius is identified with the connective cover

$$\varphi : \pi_* \mathrm{THH}(\mathbb{F}_2) \cong \mathbb{F}_2[x] \rightarrow \mathrm{THH}(\mathbb{F}_2)^{h\mu_2} \simeq \mathbb{F}_2[x^\pm].$$

On the fixed point level we must then have $\varphi : \bar{x} \mapsto b$. □

From here, the computation proceeds very similarly to that of Quigley–Shah, so we provide only a brief outline of the proof.

Theorem 3.2. *There is an equivalence*

$$\mathrm{TCR}(\mathbb{F}_2) \simeq H\mathbb{Z}_2 \oplus \Sigma^{-1}H\mathbb{Z}_2.$$

Proof Sketch. The cyclotomic Frobenius $\varphi^{hC_2\mathbb{T}}$ can be computed from the preceding proposition by analyzing the induced map on twisted homotopy fixed point spectral sequences. By definition, we have

$$\mathrm{TCR}(\mathbb{F}_2) = \mathrm{fib}(\mathrm{TCR}^-(\mathbb{F}_2) \xrightarrow{\varphi^{hC_2\mathbb{T}}-\mathrm{can}} \mathrm{TPR}(\mathbb{F}_2)).$$

The result on the homotopy Mackey functors then follows from a straightforward analysis of the long exact sequence.

To deduce the spectrum level splitting, one need only observe that the connective cover map $\tau_{\geq 0} \mathrm{TCR}(\mathbb{F}_2) \simeq H\mathbb{Z}_2 \rightarrow \mathrm{TCR}(\mathbb{F}_2)$ provides $\mathrm{TCR}(\mathbb{F}_2)$ with the structure of an $H\mathbb{Z}_2$ -module. □

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